「Lemma. Let I be a proper ideal of an integral domain R.
If a non-constant monic polynomial $P(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ satisfies
$$\overline{P}(x) = x^n + (a_{n-1}+I)x^{n-1} + \cdots + (a_1+I)x + (a_0+I) \in (R/I)[x]$$
cannot be factored two smaller degree polynomials, ($\Leftrightarrow$ irreducible)
then $P(x)$ is irreducible in $R[x]$.
Proof. Suppose that $\overline{P}(x)$ cannot be factored, and $P(x)$ is reducible.
That is, $P(x) = a(x)\cdot b(x)$ where $a(x), b(x)$ are non-constant monic polynomial.
Then, $\overline{a}(x)$ and $\overline{b}(x)$ is the factor of $\overline{P}$, it is contradiction.

「Eisenstein Criterion.
Let P be a prime ideal in the integral domain R.
Suppose that a non-constant monic polynomial
$$f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \in R[x]$$
satisfies $a_{n-1}, \cdots, a_1, a_0 \in P$ but $a_0 \notin P^2$, then f is irreducible.
Proof. Suppose that $f(x)$ is reducible, factored as $f(x) = a(x)b(x)$,
where $a(x)$ and $b(x)$ are monic, degree smaller than $f(x)$. Denote
$a(x) = x^r + \alpha_{r-1}x^{r-1} + \cdots + \alpha_1 x + \alpha_0$ and
$b(x) = x^s + \beta_{s-1}x^{s-1} + \cdots + \beta_1 x + \beta_0$. Then, By the assumption,
$$\overline{f}(x) = x^n + (a_{n-1}+P)x^{n-1} + \cdots + (a_0+P) = x^n = \overline{a}(x)\,\overline{b}(x).$$

Since P is prime, R/P is again the integral domain,
So either $\overline{\alpha_0}$ or $\overline{\beta_0}$ is zero. WLOG, let $\overline{\alpha_0} = 0$.
Then, $\overline{\alpha_1}\cdot\overline{\beta_0}$ must be zero, as the unique degree 1 coefficient of $\overline{f}$,
So $\overline{\alpha_1}$ or $\overline{\beta_0}$ is zero. if $\overline{\beta_0}$, then $\alpha_0, \beta_0 \in P \Rightarrow \alpha_0\beta_0 \in P^2$, contradiction
If $\overline{\alpha_1} = 0$, then similarly, $\overline{\beta_0} = 0$ or $\overline{\alpha_2} = 0$. Inductively, if $\overline{\alpha_{r-1}} = \cdots = \overline{\alpha_1} = \overline{\alpha_0} = 0$,
then $1\cdot\overline{\beta_0} = 0$, so, $\overline{\beta_0} = 0$, or $\beta_0 \in P$. Now, $\alpha_0\beta_0 \in P^2$, thus proved.